# LLMs for I/Os 2.0

A Functionalities and Applications Masterclass

SIOP ANNUAL CONFERENCE
NEW ORLEANS • April 29–May 2, 2026

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# Agenda

- LLMS & ChatGPT – An Overview
- Understanding the Basics
- Using LLMs Responsibly & Critically
- LLM Service & Tool Usage
- Scaling LLM Usage with an Automation Framework
- Agentic AI: Introduction & Showcase

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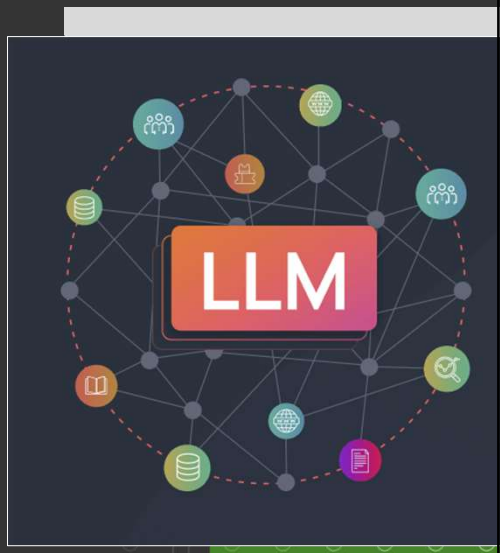
# LLMs & ChatGPT

An Overview

2 May 2026

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## Overview Agenda

1. Understanding Generative AI & GenAI platforms
2. Recent developments in the commercial platform space
3. Safety concerns & governance


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## Understanding Generative AI & GenAI platforms


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## What are Generative AI (GenAI) models?

- Complex machine learning models that uses techniques to generate output, based on a stimulus
- Models for generating text, images, and audio
- All models work by same basic principles:
  1. Represent inputs as learned numerical representations (embeddings)
  2. Align the representation across modalities (if necessary)
  3. Use a modality-specific generator to produce output

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## Understanding Embeddings

**Embedding**: A way of converting text, images, audio, or other inputs into numbers that capture meaning and patterns the AI model can use

**Natural Language Input** *The quick brown fox jumped over the lazy dog!*

↓

**Pre-processing** [*"the" "quick" "brown" "fox" "jump" "over" "the" "lazy" "dog" "!"*]

↓

**Tokenization** [791, 4062, 14198, 39935, 27002, 927, 791, 16053, 5679, 13]

↓

**Embeddings** → [0.18, -0.42, 0.07, -0.98, 0.09, ...]

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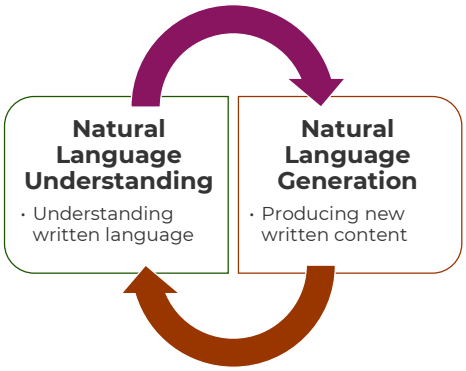
## Natural Language Processing (NLP)




- **Text Generation and Comprehension**: Crafting coherent and contextually relevant text based on input prompts.
- **Question Answering and Reasoning**: Providing informative responses and engaging in complex problem-solving.
- **Language Translation**: Converting text between languages while maintaining nuance and context.
- **Summarization**: Condensing lengthy documents into concise summaries without losing essential information.

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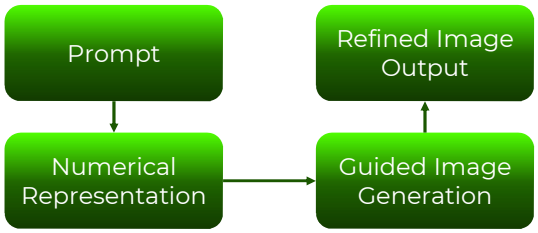
## Image Generation

Computer Vision & Image Understanding

**Layers in model ←→ levels of abstraction**

- Pixels, regions, edges
- Textures, objects
- Spatial relationships
- Style & artistic features

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## Speech Generation

Speech Recognition & Synthesis

Audio Input: Speech -> Text -> Embedding

- Words
- Tone
- Timing

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## Generative AI Platforms

- Commercial, publicly available platforms powered by artificial intelligence (AI)
  - Unified "thinking systems" powered by Large Language Models (LLMs)
- *Major LLM platforms*: ChatGPT, Claude, Gemini
- Other services
  - Chatbots
  - Browser plug-ins (e.g., Merlin)

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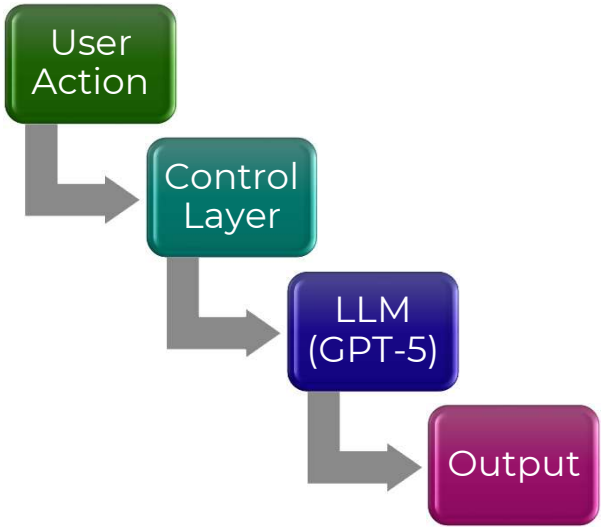
## ChatGPT as a Thinking System

**Orchestration Layer**
Executive layer that guides the flow of processing user prompts & actions

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## Recent Developments in the GenAI Platform Space


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## LLM Improvements vs. System Improvements

**LLM Improvements**

- New models with more advanced reasoning capacity
  - Model families: GPT-5 (OpenAI) | Gemini 3 (Google) | Claude Opus 4.5 (Anthropic)
- Longer context windows
  - May not always improve performance

| Factor | 2022* | April 2025 | April 2026 |
|---|---|---|---|
| Context Window Limit (Tokens) | 4,096 | (Up to) 128,000 | 272,000 – 1M+ (API) |
| Input Modality | Text only | Text, images, voice | Natively Multimodal |
| Features | Limited plugins & tools | Custom GPTs, advanced API features, agents | Agent Mode, Codex |

** 2022 was the year ChatGPT was released*


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## LLM Improvements vs. System Improvements

**System Improvements**

- Single-model chat → multi-component systems
  - routing + "thinking" variants + tool execution + retrieval + connectors + governance controls
- Agentic AI & computer use
  - Agent Mode (ChatGPT/OpenAI) | Project Mariner (Google) | Claude in Chrome (Anthropic)
  - Claude Code (Anthropic) | Codex (OpenAI)
- Emphasis on tool-mediated workflows & structured outputs
- RAG is now widespread within organizations
- Methods for reducing inference cost (e.g., mixture-of-experts)

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## Safety Concerns & Governance

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## Safety Concerns & Governance

**Enterprise GenAI: privacy, security, and governance**

- Data handling matters more than model quality alone
  - Key question went from “Can it generate good text?” to “Where does the data go, and who can access it?”
- Different account types have different privacy defaults
  - Consumer tools may use interactions to improve models unless users opt out
  - Business/enterprise offerings typically do *not* train on organizational data by default
- Organizations now need AI governance, not just AI access
  - Approved tools and account types
  - Rules for sensitive data, PII, client data, and IP
  - Human review for high-stakes outputs
- Security is now part of adoption
  - Encryption, SSO, access controls, retention policies, admin controls, auditability

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## Safety Concerns & Governance

**Agentic AI Raises New Concerns**

- Risk shifts from “bad answers” to “bad actions”
  - Traditional chat risk: inaccurate output, hallucinations
  - Agent risk: system can browse, retrieve, edit, submit, or send
- Tool access expands capabilities and ways to attack
  - Web browsing, file access, connectors to email, calendars, documents, drives
  - Terminal/code execution
  - Prompt injection becomes major concern when browsing web
- Core governance questions
  - What *tools* are enabled?
  - What *approvals* are required?
  - What *actions* need human confirmation?
  - What *logs and oversight* exist?

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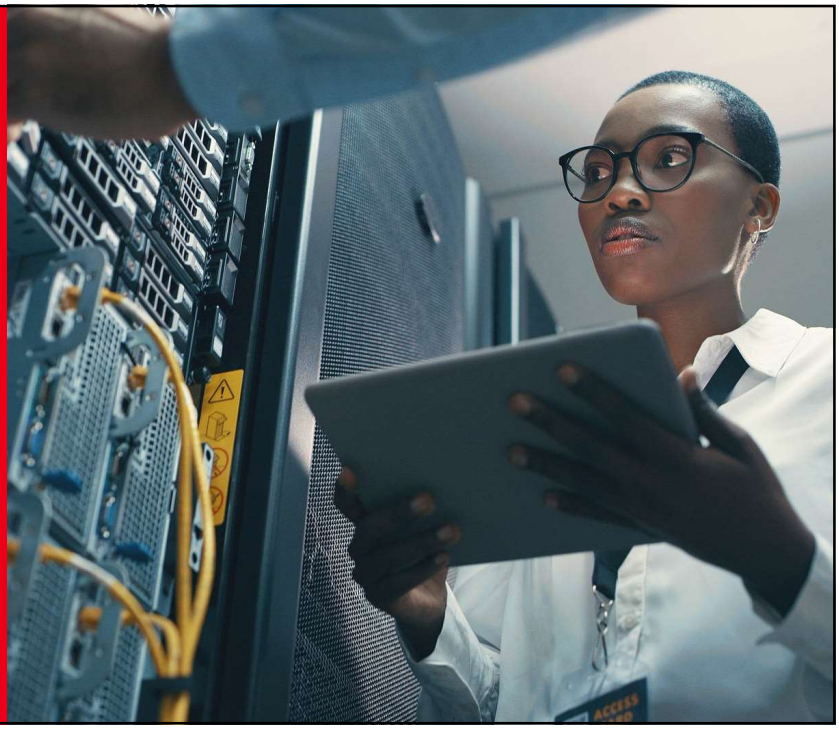
# Understanding the Basics

LLMs and Prompt Engineering

Eva Riddervold Sandberg,
Senior Market Consultant, Lic. Psychologist,
Global Science & Analytics



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## Why this matters for I/O Psychologists

- AI tools are entering everyday work
- LLMs shape communication, analysis and decisions
- Correct mental models → better use, less risk

High fluency → easy to overtrust

Fast output → changes how work is done

Broad applicability → shows up everywhere

Requires judgement → not plug-and-play

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# What is an LLM?

## Large-Language Model

- A large language model is trained to predict the next token in text
- It is pretrained on large text corpora
- It is then post-trained/fine-tuned to follow instructions and better match human preferences

**Tokenization**

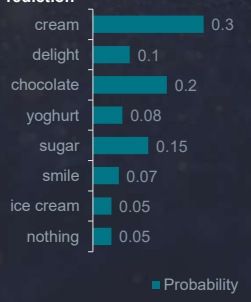
She ate a strawberry with

She ate a strawberry with

She ate a strawberry with

She ate a strawberry with

**Prediction**

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# Why it feels intelligent

- Trained on massive amounts of human text
- Learns patterns of how knowledge is expressed
- Builds associations that look like *knowledge*
- Optimized for plausible responses - not truth

**The illustion of knowledge retrieval**

Using context to narrow down the meaning

Context clues:
- Name: Michael
- Actor
- Inglourious Basterds

"You mean Michael Fassbender, the German-Irish actor"

**The model combines context, it doesn't retrieve stored facts.**

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**Stochastic parrots**

Echoes patterns, not understanding

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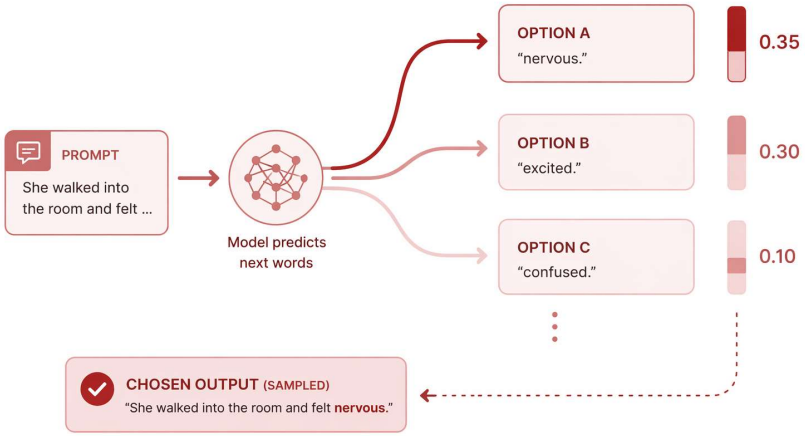
## Why outputs vary

From **multiple plausible continuations**, one is sampled each time, making the model non-deterministic.

The model samples from likely options, it doesn't pick a single correct answer.




**There is always some randomness**

**Low**:
- more predictable
- more repetitive

**High**:
- more creative
- more variable
- less reliable

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## Prompt Engineering

**Why it matters**

- The model responds to how you structure the input
- Vague prompts → vague results
- Clear prompts → more useful outputs

**Good prompts include**

- **Task**
  *What you want done*
- **Context**
  - *What the model needs to know*
- **Output format**
  *How you want the answer structured or formatted*

**Prompt A**

"Analyze these survey results."

**Prompt B**

"Summarize the main findings from these survey results."

**Prompt C**

"Summarize the main findings from these survey results, **highlight 2-3 key trends**, and **suggest possible explanations**. Present in a mix of bullet points and short paragraphs.

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## What LLMs do well

**Strong in language-based tasks**

- Summarization
- Drafting and rewriting
- Classification and extraction
- Explanation
- Brainstorming
- Synthetic first-pass analysis

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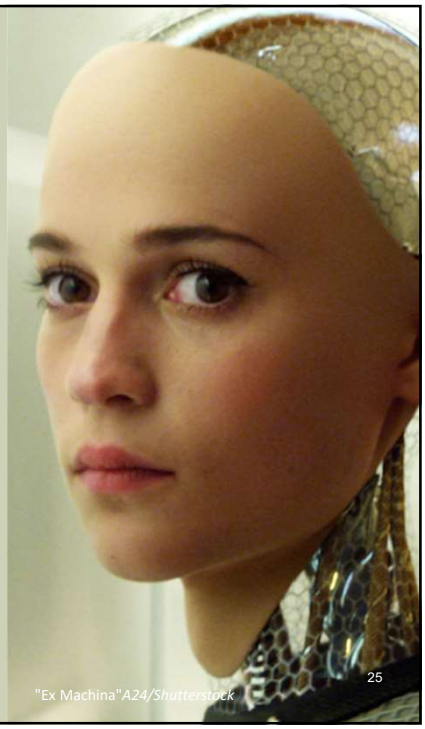
## ….and what it's not

- They are not grounded in real-time truth
- They can produce plausible but false statements
- They are not sentient; they do not have beliefs, intent, or human understanding
- They are not complete systems

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# Using LLMs Responsibly

Responsibility & Applying a Critical Eye

5/2/2026

Luke Camden

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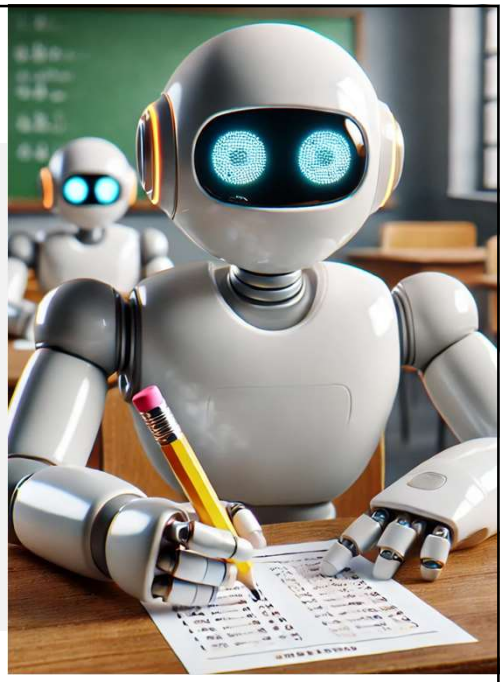
## A powerful tool for I/O work

What If I Told You...

- AI algorithms improved the accuracy of personality assessments by 15% compared to traditional methods (Smith & Johnson, 2019).
- AI-driven testing can reduce the assessment time by 30% compared to traditional methods, while maintaining similar levels of accuracy (Johnson & Thompson, 2018).
- AI algorithms can reduce bias in personality assessments by 25% compared to human evaluators (Wang & Chen, 2021)



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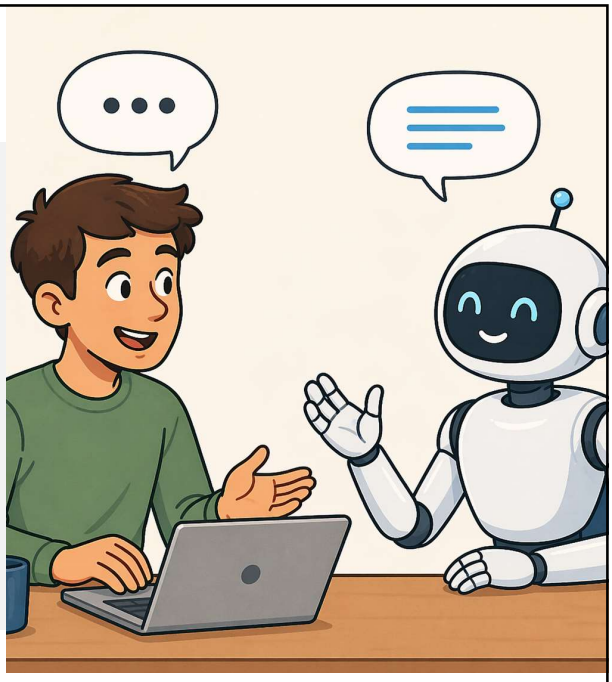
## Why does it matter?

The effects of improper use

- LLM output can be fluent, specific, and wrong.
- Credibility of tone is not evidence of accuracy.
- We stake our own reputation on results when we use LLMs.
- Responsible use starts with matching the tool to the task.



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## LLMs are tools like any other

Match the tool to the job

- Powerful for drafting, summarizing, and ideation
  - Not appropriate for every task
  - Use them to support judgment, not replace it
  - Avoid the “hammer-and-nail” trap
- Don’t let everything become a nail

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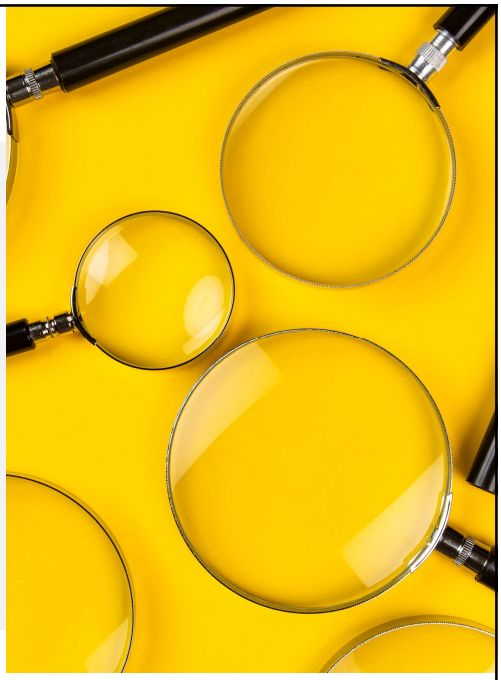
## What to look for when reviewing LLM generated content

Signs of inaccuracy

- Hallucinations
  - Information that sounds credible but is inaccurate or made up.
- Source & Credibility Bias
  - Sources identified may only be included to support a claim without proper review for credibility
- Over confidence or unsupported claims
  - language that sounds more certain or definitive without providing sufficient evidence.
- Circular Language
  - Polished wording that repeats or rephrases an idea without clearly explaining it.



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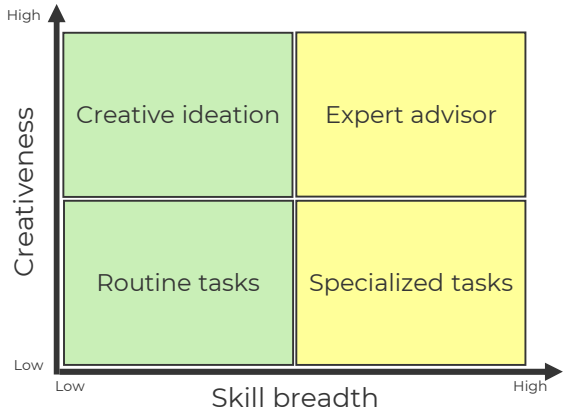
## Understanding model limitations

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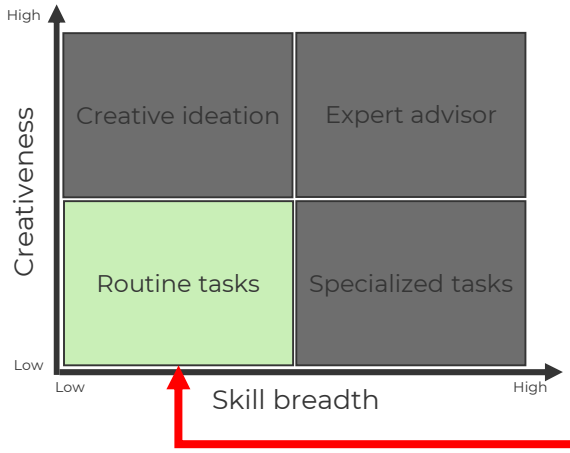
## Understanding model limitations




**Routine Tasks:** Requires the model to complete a simple task which requires little to no creativity and is limited to a small number of skill domains requiring little context.

**Examples:**

- Summarizing provided text
- Boiler plate scripting
- Reviewing text or code for errors

**Risk**: Minimal

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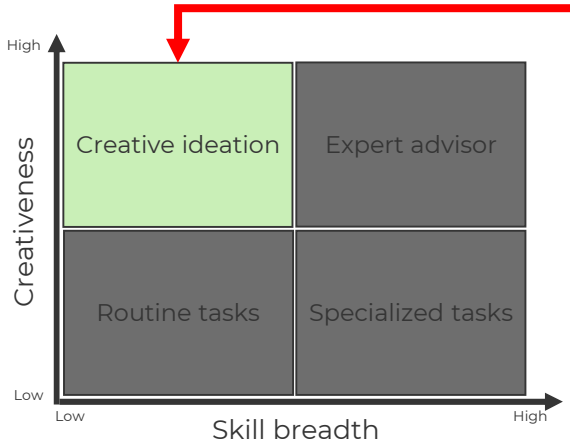
## Understanding model limitations




**Creative ideation:** Requires the model to generate creative ideas which can be expanded upon or applied in a limited manner.
**Examples:**
- Refining text that has already been generated
- Brainstorm ideas

**Risk**: Minimal

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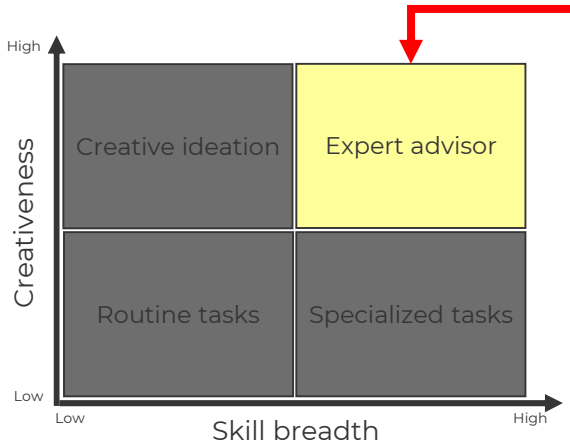
## Understanding model limitations




**Expert advisor:** Requires the model to use creativity to connect multiple skill domains to find a solution.
**Examples:**
- Designing a talent strategy
- Creating a plan for a new application
- Creating an organizational policy

**Risk:** Connecting multiple skill domains in a creative way can require contextual or ethical understanding that the model may lack.

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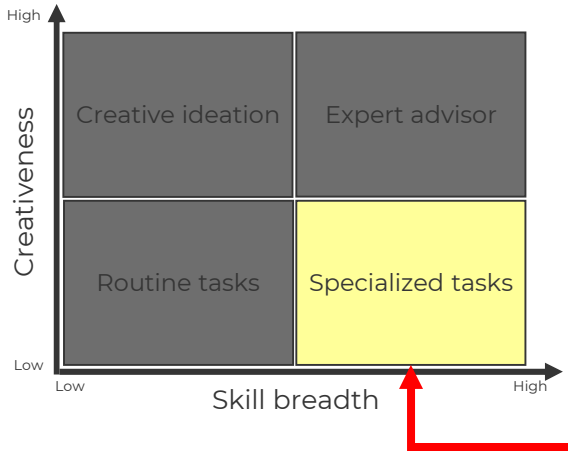
## Play to the model's strength




**Specialized task:**
Requires the model to use complete a task in a very specific way that requires multiple skill areas.
**Examples:**
- Generating citations
- Reviewing the quality of presented research
- Interpreting complex analysis

**Risk:** if not properly supplied with information the model may hallucinate to help it achieve its goals.

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## Considerations for using LLMs

Best Practices

**Treat ChatGPT as a knowledgeable and eager, albeit inexperienced and sometimes careless, assistant**

- Clear, specific prompts
  - ❌ "Tell me about remote work"
  - ✅ "What does recent research say about the impact of remote work on employee productivity and satisfaction?"
- Save conversations & ask for links to be printed
  - ✅ After the search is concluded: "For all sources cited, print out the web page name and URL of each source without markdown formatting."
- Steer towards high-quality & reputable sources
  - ✅ Tell it which sources to focus on
  - ✅ Tell it which sources to avoid
- Avoid the black box as much as possible
  - ✅ Document when and how you use ChatGPT web searching for any work or research

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## Considerations for using LLMs

Best Practices

- Be critical
  - "Do I really need to use an LLM here? Does the benefit outweigh the cost of not doing the actual search myself?"
    - ✅ Review all sources for accuracy of claims
    - ✅ Critically evaluate the claims being made
  - Best use case may be to identify & categorize sources, which you review manually

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# LLM service and tool usage

WHAT HAPPENS AFTER YOU HIT "ENTER"?

Bao Ho

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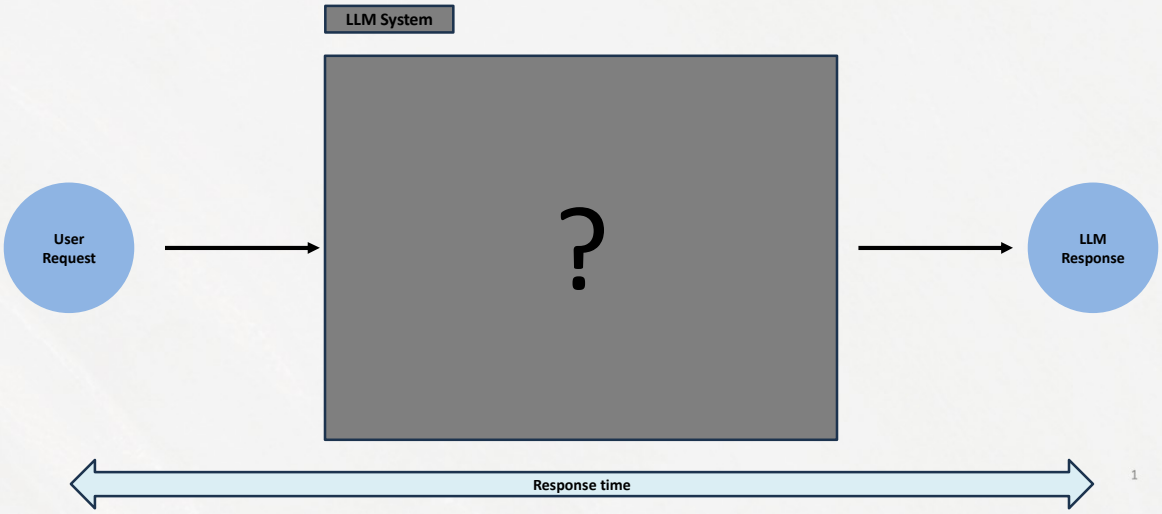
# How LLMs appear from the outside

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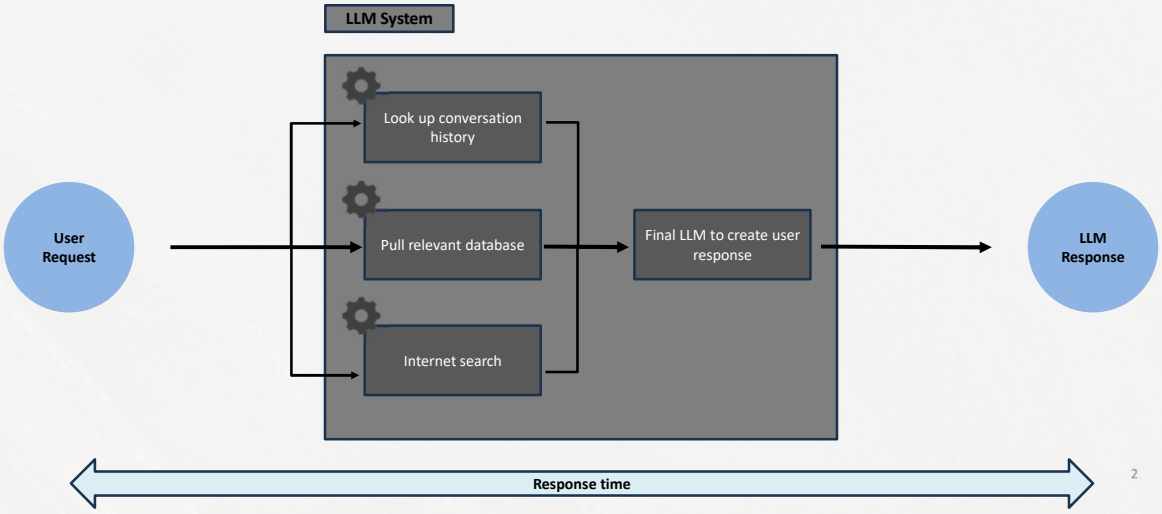
# Inside the LLMs service

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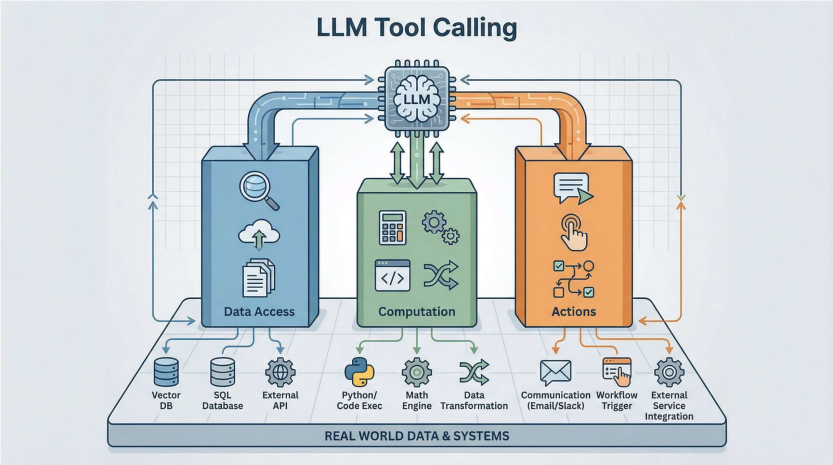
# Why tools calling?




**LLM can call external tool:**

- Retrieval (documents, databases)
- Web (real-time info)
- Code (Python execution

**Why this matters?**

- More accurate
- More up-to-date
- Can compute, not just predict

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# Tools usage example

From engagement survey, analyze trends and identify key driver of turnover intention

**1. Input understand**

User provides survey data and research questions

**2. Decision making**

LLM decide to use analysis tool (R, Python)

**3. Data processing**

Clean and structure survey data

**4. Analysis execution**

Run analysis to find key driver

**5. Output and insight**

Return charts + explain insights

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# How to get better results from LLMs

**Clear prompt**

- Specify the task clearly
- Include context (data, goal, audience)
- Avoid vague instruction

**Use structure**

- Ask for output format (e.g., JSON, tables, bullet points)
- Makes results easier to validate/reuse
- Helps with automation

**Break it down**

- Split complex tasks into step
- Improve accuracy
- Easier to debug results

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# Scaling LLMs with Automation Framework:
## From Web Interfaces to Automation

Yizhen Egyn Zhu
Machine Learning Scientist
Amazon - Intelligent Talent Acquisition

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# Scaling Problem with Web Interfaces

- How would you do the following with ChatGPT?
  - Generating assessment items across 50 jobs × 10 competencies
  - Processing resumes from thousands of applicants
  - Extract data from thousands of interview notes

Each prompt requires unique contextual information

Manual copy-paste is time consuming and error-prone

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# Capabilities of Automation

Automated Context Injection
- Automatically insert context into prompt templates
- Like mail merge for AI prompt

Batch Processing
- Process multiple requests in parallel
- Like sending 1,000 emails at once instead of one-by-one

Automated Output Structuring
- Get organized data with specified fields instead of plain text
- Like receiving a spreadsheet instead of randomly structured text

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# Why Does I-O Need LLM Automation?

New Research Opportunities
- Allows large-scale studies utilizing LLMs
- One researcher can run thousands of prompts in a minute

Improved Quality and Consistency
- Structured outputs allows standardized post-hoc analysis
- Prompt templates and code makes research reproducible

Saves Time for Higher-value Work
- Reproducible code allows fast prompt/model iterations and comparisons
- Researcher can focus on validation and interpretation

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# Demo – LangChain

LangChain: A popular framework for LLM automation
Use case: Generate interview questions

**You don't need to code along today** – just follow the logic
Complete notebook will be available on GitHub after the presentation for you to explore at your own pace

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# Step 1 – Set Up

```
# Install and import LangChain
#!pip install "langchain>=0.3,<0.4" "langchain-community>=0.3,<0.4"   # only needs to be done once
from langchain_core.prompts import PromptTemplate
from langchain_core.output_parsers import PydanticOutputParser
from langchain_community.chat_models import ChatOllama
from pydantic import BaseModel, Field
from typing import List
```

✓ 0.6s

- Using an open-source Llama model for demo
  - In practice, you can use your own preferred models
  - E.g., GPT (ChatOpenAI) or Claude (ChatBedrock)

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# Step 2 – Create Prompt Template

```
# Define a prompt template with placeholders
template = """You are an Industrial-Organizational psychologist creating interview questions.

Job Title: {job_title}
Competency: {competency}
Competency Definition: {competency_definition}

Generate 3 behavioral interview questions that assess this competency
for this role.

{format_instructions}
"""

# Define input variables according to the curly brackets in prompt
# Note. format_instructions is set via partial_variables so the parser's
# instructions are automatically injected into every prompt
prompt = PromptTemplate(
    input_variables=["job_title", "competency", "competency_definition"],
    partial_variables={"format_instructions": ""},  # placeholder, will be filled after parser is created
    template=template
)
```

✓ 0.0s

amazon

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# Step 3 – Specify Output Structure

```python
# Define what structure we want the LLM to return
class InterviewQuestion(BaseModel):
    question: str = Field(description="The interview question text")
    explanation: str = Field(description="Why this question assesses the competency")

class QuestionSet(BaseModel):
    questions: List[InterviewQuestion]

# Create a parser that enforces this structure
parser = PydanticOutputParser(pydantic_object=QuestionSet)
```

✓ 0.0s

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# Step 4 – Construct the Chain

```python
# Initialize the LLM (requires downloading and installing Ollama: https://ollama.com)
#!ollama pull llama3    # Download the model (only needs to be done once)

llm = ChatOllama(model="llama3", temperature=0.7)

# Inject the parser's format instructions into the prompt
prompt_with_instructions = prompt.partial(format_instructions=parser.get_format_instructions())

# Connect all the pieces together
chain = prompt_with_instructions | llm | parser
```

✓ 0.0s

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# Step 5 – Process Jobs

```python
# Your data: multiple jobs and competencies
jobs_data = [
    {"job_title": "Software Engineer",
     "competency": "Problem Solving",
     "competency_definition": "Ability to analyze complex problems."},
    {"job_title": "Product Manager",
     "competency": "Stakeholder Management",
     "competency_definition": "Ability to manage diverse stakeholder needs."},
    {"job_title": "Data Scientist",
     "competency": "Analytical Thinking",
     "competency_definition": "Ability to interpret complex data patterns."},
    # In practice: 500+ entries from your job analysis database
] # This can also be stored and read from a spreadsheet (e.g., csv file)

# Process all at once in parallel
results = chain.batch(jobs_data, config={"max_concurrency": 3})

# Export to spreadsheet
import pandas as pd
df = pd.DataFrame([
    {"job": d["job_title"], "competency": d["competency"],
     "question": q.question, "explanation": q.explanation}
    for d, r in zip(jobs_data, results)
    for q in r.questions
])
df.to_csv("interview_questions.csv", index=False)
```

✓ 19.7s

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# Output csv

| job | competency | question | explanation |
|---|---|---|---|
| Software Engineer | Problem Solving | Tell me about a time when you were faced with a complex technical problem in your previous software development project. How did you approach it, and what was the outcome? | This question assesses the candidate's ability to analyze complex problems by understanding how they approached a difficult issue in their previous experience. |
| Software Engineer | Problem Solving | Can you describe a situation where you had to debug an issue that was causing errors in your code? How did you go about identifying and resolving the problem? | This question evaluates the candidate's ability to analyze complex problems by understanding how they identified and resolved an error-causing issue. |
| Software Engineer | Problem Solving | Tell me about a project where you had to integrate multiple software components to achieve a specific goal. How did you handle any unexpected issues that arose during integration, and what was the outcome? | This question assesses the candidate's ability to analyze complex problems by understanding how they handled unexpected issues during an integration project. |
| Product Manager | Stakeholder Management | Tell me about a time when you had to manage competing priorities from multiple stakeholders in a product development project. How did you handle it? | This question assesses the candidate's ability to prioritize and manage diverse stakeholder needs, ensuring that all parties are satisfied with the outcome. |
| Product Manager | Stakeholder Management | Can you describe a situation where you had to negotiate with stakeholders who had differing opinions on what features should be included in a product? What was the outcome? | This question evaluates the candidate's ability to navigate stakeholder expectations and find common ground, ultimately leading to a successful product launch. |
| Product Manager | Stakeholder Management | Tell me about a project where you had to balance the needs of various stakeholders, such as customers, engineers, and executives. How did you ensure that everyone's concerns were addressed? | This question assesses the candidate's ability to effectively manage stakeholder expectations across different levels and departments, ultimately delivering a product that meets all stakeholders' needs. |
| Data Scientist | Analytical Thinking | Can you describe a time when you analyzed a complex data set and identified patterns or trends? How did you determine what was important and what wasn't? | This question assesses the candidate's ability to interpret complex data patterns, looking for specific examples of how they prioritized and extracted insights from large datasets. |
| Data Scientist | Analytical Thinking | Tell me about a situation where you had to untangle conflicting data points or contradictory findings. How did you resolve the issue? | This question evaluates the candidate's analytical thinking skills, focusing on their ability to identify and address inconsistencies in complex data sets. |
| Data Scientist | Analytical Thinking | Describe a project where you used statistical methods or machine learning algorithms to extract insights from large datasets. What did you discover, and how did you present your findings? | This question examines the candidate's analytical thinking skills, looking for evidence of their ability to apply statistical or algorithmic techniques to complex data sets and effectively communicate their results. |

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# Next Steps

- **GitHub notebook** will be shared after this session
  - You can run the notebook locally if you already have Python
  - Else, Google Collab is a good starting place
- **You don't need to be a programmer to use this:**
  - You can modify the template and data for your own research
  - You can even ask AI to modify the template, format your data, troubleshoot errors (:
  - Start small: try with 5-10 jobs first, then scale up
  - Focus on the prompt and data—the code handles the rest

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# Appendix: Beyond Batch Processing

What's Covered Today: Batch Processing
- Good for pre-defined tasks
- E.g., item generation, structured data extraction, free text scoring

Next Frontier: Agentic Workflows
- AI agents can plan, reason, and make (semi-) autonomous decisions
- Example framework: LangGraph, an extension of LangChain
- E.g., An agent that generates/accepts interview questions, search in a predefined pool of dimensions to evaluate quality, refines questions, iterates with user until satisfied

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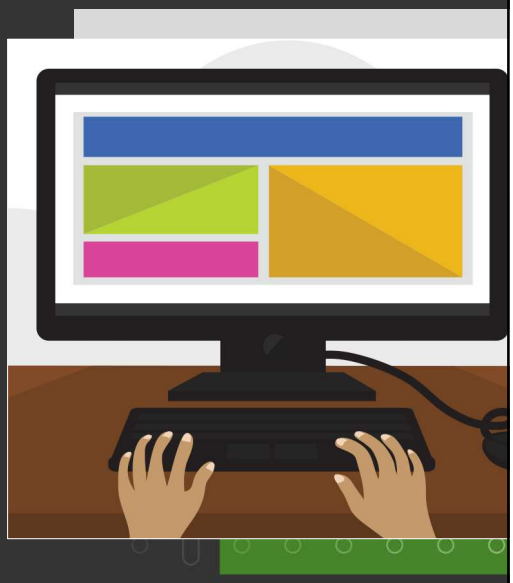
# Agentic AI: An Introduction

Understanding Agentic Systems

2 May 2026



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## Setting the Scene: What is Agentic AI?

AI **Chatbot** v.s. AI **Agent**

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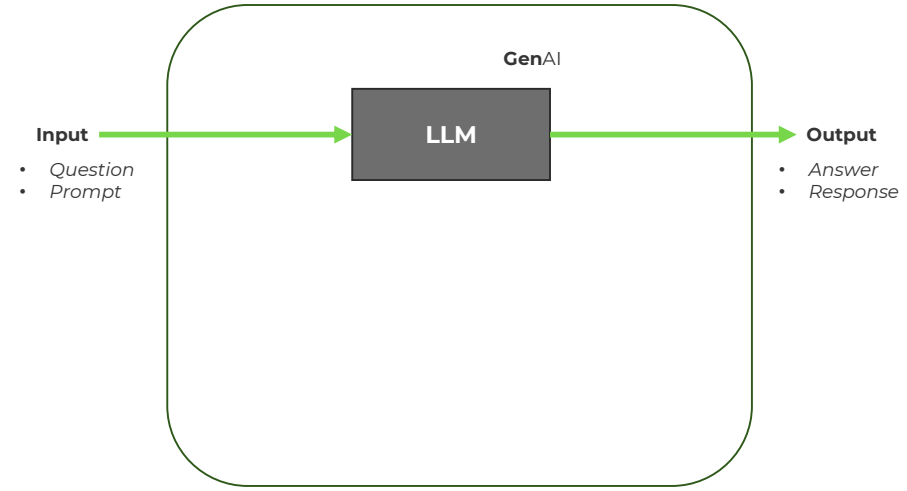
## Setting the Scene: What is Agentic AI?

AI **Chatbot**





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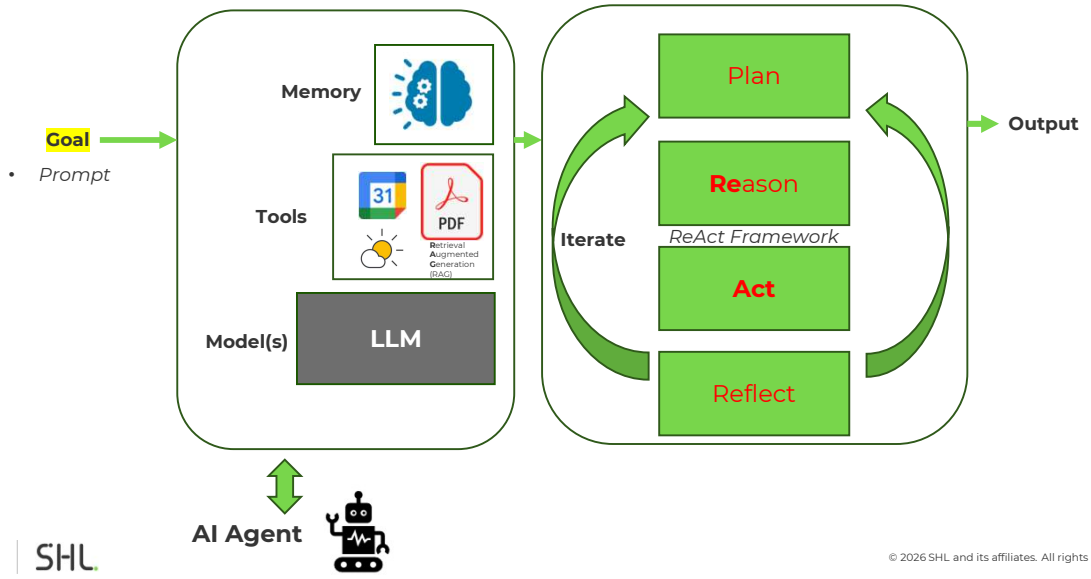
## Setting the Scene: What is Agentic AI?

AI **Agent**





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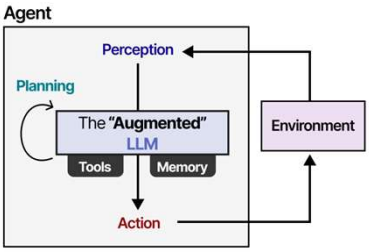
## Setting the Scene: What is Agentic AI?

*An Agent is a system that leverages an AI model to **interact** with its environment in order to achieve a user-defined **objective**. It combines **reasoning**, **planning**, and the **execution** of actions (often via external tools) to fulfil tasks.*

| AI Chatbot | AI Agent |
|---|---|
| **Chats** | **Acts** |
| **Reactive** | **Proactive** |
| Responds to what you type | Plans what to do and executes |
| Follows preset flows | Plans next steps |
| Serves canned answers | Learns from memory |

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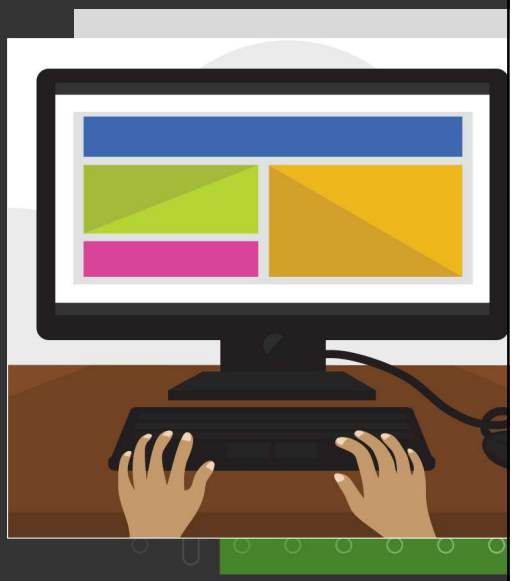
## Agentic AI: Using ChatGPT's Codex

Best Practices & Demonstration

2 May 2026



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## What is Codex?

- Codex is an **AI Coding Agent**
  - AI systems that have elevated permissions to read & write computer files
  - Assist with coding & other complex tasks
- Other major coding agents
  - Claude Code
  - Codex
  - Cursor
- Methods of use
  - IDE extensions (e.g., VS Code)
  - CLI (Command-line interface)
  - Dedicated application

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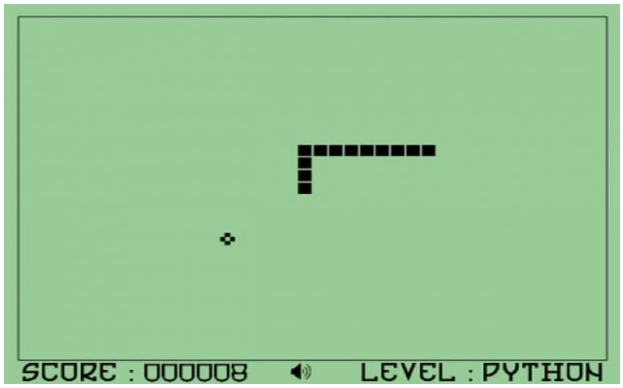
## Live Example: Classic Snake

Let's make a game!

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## BEST PRACTICES (Before you start)

- *Best Practice Links*: Claude Code - Claude Code Docs ; Codex | OpenAI Developers
- Getting set up
  - Using the VS Code extension for demonstration
- Review configuration file
  - *Universal defaults*: "~/.codex/config.toml" (Config basics – Codex | OpenAI Developers)
  - *Repo-specific behavior*: ".codex/config.toml"
- Constrain access
  - Use config files, containerization, and other techniques to limit the scope of what the agent can access & permissions it has
  - Be conservative with the access & permissions you give
  - Codex: Key settings in config files: *approval* (permissions for what commands can be executed without user permission at run time) and *sandbox* (which directories the agent has read/write access to)
- Version control
  - Use git or other version control to track what it is doing & save files elsewhere

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## BEST PRACTICES (Explore, Plan, Implement)

- You, the human, need to have a solid understanding of project parameters: resources, objectives, strategy, verification, and deliverables.
- Aggregate all relevant information (including links to other sources) in project directory
- For complex tasks, create a *design document*
  - Iterate with ChatGPT to ensure you cover everything major (esp. for technical tasks that are outside your current range of knowledge, to reduce unknown unknowns)
  - Can ask ChatGPT to "interview" you to get a full understanding of the project
- **Explore & Plan**:
  - (Depending on the project, this may have multiple steps or just go straight to creating a Plan.)
  - Enter "Plan Mode" and have the agent read files and ask questions.
  - Have the agent come up with a detailed implementation plan and review it carefully. Make changes as necessary. (As the human in charge, this is one of the most important steps to get right.)
- **Implement**: Once the plan is set, have the agent implement the plan.
  - Keep an eye on the kind of work it is doing; stop the agent if it is going astray.
  - Empower it to verify its own work (highest-leverage thing you can do). Provide clear verification criteria and ensure it is being met.

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## BEST PRACTICES (Getting started)

- Initial prompt (usually given when asking it to explore & create a plan): Important features
  - **Goal**: What are you trying to change or build?
  - **Context**: Which files, directories, examples, and sources matter for the task? (Should all be easily accessible)
  - **Constraints**; What standards, architecture, safety requirements, or ethics should the agent follow?
  - **Done when**: What conditions or tests will signal when the agent is completed with the task?
- Reasoning level:
  - *Low*: Small, fast, well-scoped tasks
  - *Medium/High*: Complex changes or debugging
  - *Extra High*: Long, agentic, reasoning-heavy tasks

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## BEST PRACTICES (General Patterns & Principles)

- MD files (Codex: AGENTS.md, ***etc...***) can be used to create easy-to-reference guidance for agents, executive plans that can be referenced and refreshed/updated, etc.
  - Instruction files should be short, practical, and behavior-focused
- Manage the context window—the most precious resource
  - As context window fills, performance degrades (forgets earlier commands, increased confusion/noise, etc.)
- Treat the agent like a teammate—not autocomplete! You define the goal, context, constraints, and completion status.
- Keep durable instructions in the repository (e.g., AGENTS.md or CLAUDE.md), not the chat.
  - Keeps things from being lost in the context window
  - “Central plan” to which agent and subroutines can refer
- Be concrete about where the agent should look and what pattern(s) it should follow
  - Useful to use relative paths for projects

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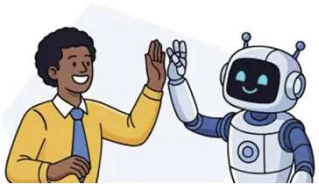
## CODEX PROJECTS & USE CASES

- Coding
  - **Pair programming**
    - Can and should ask for input, but should always critically review plans and make changes
    - *Examples*: R packages & Shiny apps, other apps
  - Consolidation & database mapping
    - *Examples*: Mapping the RDW via many pulls & small extracts
  - Suggesting best practices & approaches
- Information
  - Generating annotated bibliographies
  - Research synthesis and evidence tables
  - Organize & structuralize unstructured notes, transcripts, etc.



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# Q&A Resources

- Q&A
- Slides available on **Whova** (PDF)
- All materials (slides, code, and supplementary resources) found on **GitHub repo** here:

karimhbadr1/SIOP_2026_LLMs-for-IOs-2.0

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